Worksheet 1.1

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1. **Aim/Overview of the practical:**
   Use Hugging Face's pre-trained T5 model to summarize news articles into concise summaries.

Example-1

Coding:

```
pip install transformers
from transformers import pipeline
summarizer_pipeline = pipeline("summarization", model="facebook/bart-large-cnn")
text ="Chandigarh University is a leading private university located in Mohali, Punjab, India. Established
in 2012, the university has rapidly grown into one of India's most prominent institutions for higher
education, known for its modern infrastructure, industry-oriented curriculum, and strong focus on
research, innovation, and global exposure. Recognized by the University Grants Commission (UGC) and
accredited with an A+ grade by NAAC, Chandigarh University has consistently ranked among the top
private universities in India. The university offers a wide range of undergraduate, postgraduate, and
doctoral programs across disciplines such as Engineering and Technology, Management, Computer
Applications, Sciences, Arts and Humanities, Commerce, Law, Architecture, Design, Hotel
Management, Pharmacy, and Applied Health Sciences. Its academic structure is designed to blend
theoretical knowledge with practical skills, ensuring that students are well-prepared for professional
challenges. Chandigarh University places strong emphasis on outcome-based education, continuous
assessment, and experiential learning through labs, projects, internships, and industry collaborations.
One of the major strengths of Chandigarh University is its robust industry interface. The university has
established partnerships with leading national and international companies, enabling students to gain
hands-on experience through internships, live projects, certifications, and industrial training. This
industry-driven approach has contributed to impressive placement records, with students securing
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roles in reputed organizations across IT, engineering, management, research, and consulting sectors. The dedicated placement and career development cell works closely with students to enhance employability through training in technical skills, communication, aptitude, and personality development. Chandigarh University is also known for its strong research and innovation ecosystem. The university promotes interdisciplinary research through specialized research centers, funding support, incubation facilities, and collaboration with global universities and organizations. Faculty members and students actively contribute to patents, publications, startups, and social impact projects. The university's innovation and entrepreneurship initiatives encourage students to transform ideas into viable ventures. The campus life at Chandigarh University is vibrant and inclusive, offering a balanced mix of academics, cultural activities, sports, and student-led clubs. The sprawling, well-planned campus features modern classrooms, advanced laboratories, libraries, hostels, sports complexes, and recreational facilities. Students from diverse cultural, regional, and international backgrounds create a dynamic learning environment that promotes cultural exchange and holistic development."

```
summary= summarizer_pipeline(text, min_length=50, max_length=100)
print(summary)
```

OUTPUT:

```
[{'summary_text': 'Chandigarh University is a leading private university located in
Mohali, Punjab, India. Established in 2012, the university has rapidly grown into one
of India's most prominent institutions for higher education. It is known for its
modern infrastructure, industry-oriented curriculum, and strong focus on research,
innovation, and global exposure. The university offers a wide range of undergraduate,
postgraduate, and doctoral programs.'}]
```

Example-2

Coding:

```
pip install transformers torch sentencepiece
from transformers import T5Tokenizer, T5ForConditionalGeneration

def summarize_text(text):
 model_name = "t5-small"
 tokenizer = T5Tokenizer.from_pretrained(model_name)
 model = T5ForConditionalGeneration.from_pretrained(model_name)

 input_text = "summarize: " + text
 inputs = tokenizer.encode(
  input_text,
  return_tensors="pt",
  max_length =512,
  truncation=True
 )

 summary_ids = model.generate(
  inputs,
  max_length = 120,
  min_length=40,
  length_penalty=2.0,
  num_beams=4,
  early_stopping=True
 )

 #Decode Summary
 summary = tokenizer.decode(summary_ids[0], skip_special_tokens=True)
 return summary

if __name__ == "__main__":
 print("------ TEXT SUMMARIZATION USING T5 ------\n")
 text = """Biology is the foundation of understanding life in all its forms. It connects humans with
nature and provides solutions to many global challenges such as disease control, food security, and
environmental sustainability. By studying biology, we gain not only scientific knowledge but also a
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deeper respect for life.In today's rapidly changing world, biology continues to evolve, offering new discoveries and innovations that shape the future of humanity. Whether in classrooms, laboratories, hospitals, or natural ecosystems, biology remains a powerful tool for improving the quality of life and ensuring the survival of our planet."""
 summary = summarize_text(text)
 print("\n------ SUMMARY ------\n")
 print(summary)
```

OUTPUT:-

```
------ TEXT SUMMARIZATION USING T5 ------


------ SUMMARY ------

biology is the foundation of understanding life in all its forms. it connects humans with
nature and provides solutions to many global challenges. biology continues to evolve,
offering new discoveries and innovations that shape the future of humanity.
```

3. Learning outcomes (What I have learnt):

- Understand the fundamentals of Natural Language Processing (NLP) and abstractive text summarization.
- Load and use Hugging Face's pre-trained T5 model for text summarization tasks.
- Preprocess and tokenize news article text suitable for transformer-based models.
- Generate concise summaries from long news articles using the T5 summarization pipeline.
- Evaluate the quality of generated summaries based on coherence, relevance, and conciseness.